



Exploring the Algarrobo Decline in the Pómac Forest: Unraveling the Relationship between the Endomicrobiome of *Neltuma pallida* and *Enallodiplosis discordis*

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Abstract

A progressive decline in populations of *Neltuma pallida* (Humb. & Bonpl. ex Willd.) C.E. Hughes & G.P. Lewis (algarrobo) has been documented in Peru. While this phenomenon has been primarily attributed to herbivorous insect attacks, it has been proposed that disease severity and symptoms are exacerbated by infection with opportunistic microorganisms. *Enallodiplosis discordis* has been identified as the main pest of *N. pallida*; with larvae producing chlorotic halos on leaflets that have been suggested to involve microbial activity. In this study, we explored the relationship between the bacterial endomicrobiome of *N. pallida* branches and the microbiota of *E. discordis* larvae in the Pómac Forest Historical Sanctuary, a protected, *Neltuma* forests. We sampled branches from asymptomatic, incipiently diseased, late-stage symptomatic trees, and *E. discordis* larvae. Bacterial DNA was extracted, and 16 S rRNA gene libraries were prepared and sequenced using ONT long-read sequencing. Our results revealed that *N. pallida* branch-associated samples, regardless of health status, exhibited low bacterial diversity and were dominated by host sequences, indicating organellar carryover and insufficient detection of true bacterial endophytes in branch tissues. To assess potential bacterial overlap between the host and the pest, we characterized the microbiota of *E. discordis* larvae, which exhibited greater bacterial diversity, with limited overlap restricted to low- to moderate-abundance bacterial families such as Acidithiobacillaceae and Erwiniaceae. These exploratory analyses suggest that the observed decline symptoms may not be directly mediated by a specific high-abundance component of the bacterial community in the stems and contribute to our understanding of the interplay between biotic factors influencing algarrobo decline.

Keywords *Neltuma pallida* · *Enallodiplosis discordis* · 16S rRNA · Carob tree · Gut microbiota · Endophytes

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Introduction

Trees of the genus *Neltuma* previously known as *Prosopis* [1], commonly known as algarrobo in Peru, are distributed across the arid and desert zones of the seasonally dry tropical forest ecosystem in the country's northwest, with *Neltuma pallida* being the dominant species [2, 3]. Algarrobo is considered one of the most ecologically and economically valuable tree species in this ecosystem [2–4]. However, its populations are increasingly threatened by land-use change, illegal logging, and the rising incidence of pests and diseases [3]. Currently, it has been stated that the North Peruvian dry forest ecosystem could disappear within the next few years [5], a phenomenon termed as “algarrobo decline”. As the etiology of the non-anthropogenic factors driving this

decline remains poorly understood, identifying its potential causes is critical [5].

Algarrobo tree mortality has been mainly attributed to biotic factors, with insects from the families Tortricidae, Cecidomyiidae, and Noctuidae identified as the principal pests [6]. In 2016, eight insect species were recorded as associated with algarrobo damage [2]; however, it was not until 2019 that the dipteran *Enallodiplosis discordis* was identified as the most significant pest affecting this species in Peru [2]. *E. discordis* is present year-round on infested trees and is considered a prolific pest due to its life cycle. In its larval stage, it attaches to the leaflets of seedlings and mature trees, causing discoloration, wilting, and defoliation [2, 6]. While insect damage is often the most conspicuous symptom, insects frequently interact with opportunistic microorganisms such as bacteria and fungi [6]. Accordingly, in *N. pallida*, larvae of *E. discordis* have been observed to produce chlorotic rings at their attachment sites on foliar tissue, potentially linked to filamentous fungi or bacteria [6]. Some diseases are caused by the presence of endomicroorganisms capable of inducing dysbiosis in host tissues [7]. This observation has led to the hypothesis that certain bacterial species may play an important role in the wilting mechanism observed in *N. pallida* [6, 7].

Several studies have shown that, beyond genetic factors, plant resistance and tolerance to disease are strongly influenced by the presence of endophytic microorganisms that naturally inhabit plant tissues [8]. These microorganisms can modulate host–pathogen interactions through the production of antimicrobial compounds or by competing with pathogens, thereby enhancing host tolerance [7–9]. The gut

microbiota of phytophagous insects plays essential roles in counteracting plant defensive metabolites and reducing their toxicity [10–12].

Recent advances in molecular technologies have greatly enhanced our ability to assess the true breadth of microbial biodiversity [12, 13]. Among these, metagenomics has enabled broad access to prokaryotic information from environmental samples, regardless of culturability [12, 13]. Metabarcoding allows for the reconstruction of microbial community taxonomy from environmental or host-associated samples—such as endophytes and endomicrobiomes—through the analysis of specific marker genes like 16 S rRNA for bacterial identification [12]. Here, we used full-length 16 S rRNA sequencing to profile *N. pallida* branch endophytes and the larval microbiota of *E. discordis*, comparing diversity, composition, and host–pest microbiome overlap as a baseline for microbial surveillance of algarrobo decline.

Methods

Our workflow encompassed field sampling, surface sterilization and DNA extraction, 16 S rRNA library preparation and sequencing, and downstream taxonomic and diversity analyses; Fig. 1 provides a visual summary, with full procedures in the following subsections.

Sample Collection

Sampling was conducted in the Pómac Forest Historical Sanctuary (SHBP) in Lambayeque, Peru. Three categories

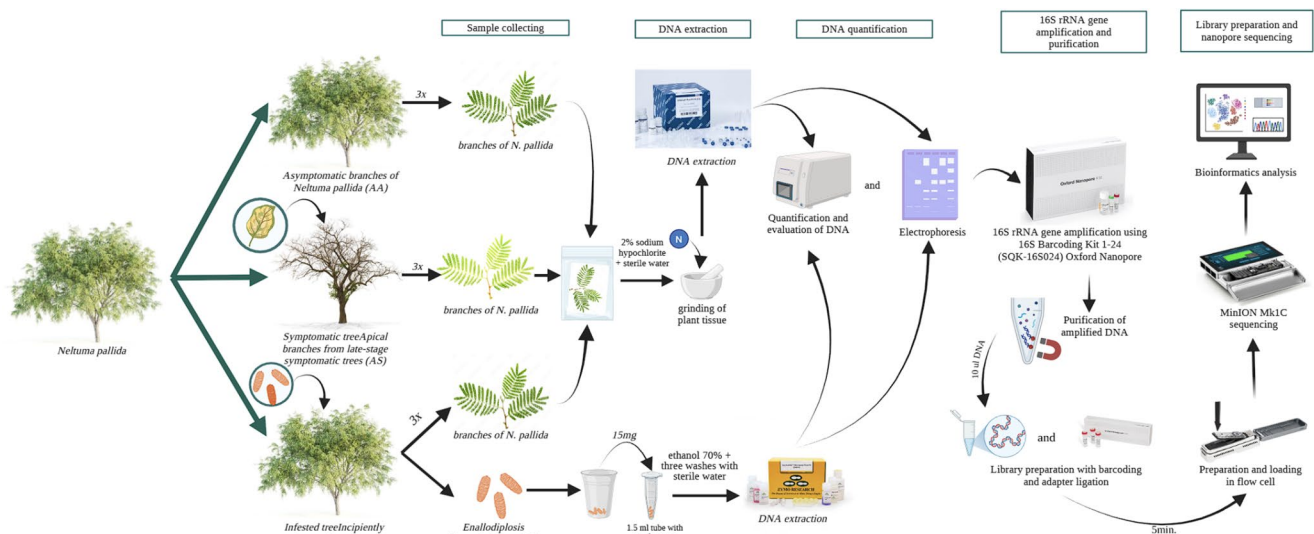


Fig. 1 Visual summary of the workflow, including sampling, surface sterilization, DNA extraction, 16 S rRNA library preparation, sequencing, and downstream analyses

of *N. pallida* trees were selected: visually healthy (asymptomatic) trees (AA), characterized by dense foliage, green leaflets, and absence of *E. discordis* larvae; incipiently diseased trees (AI), exhibiting yellowing leaves and visible *E. discordis* larvae on the leaflets; and late-stage symptomatic trees (AS), nearly defoliated, with foliage restricted to a few apical branches due to long-standing progressive wilting (Fig. 1). *E. discordis* larvae (ED) from the incipiently diseased trees (AI) were also collected.

Apical branches, consisting of the proximal stem to the leaf lamina (excluding leaflets), were collected from three asymptomatic (AA), incipiently diseased (AI), and late-stage symptomatic (AS) trees (three branches per tree; a total of nine branch samples per health category; $n=3$ trees [biological replicates] per category). Each branch was cut, rinsed with sterile water, and surface-sterilized in the field with 70% ethanol. Larvae (ED) were obtained from the same AI trees by gently detaching them from infested leaflets borne on the same branches, with one larval subsample per branch, yielding three larval subsamples per tree (nine in total). Samples (branches and larvae) were collected from three different branches within each tree canopy to maximize representativeness, coded, and transported on ice to the Genomics and Bioinformatics Laboratory for Biodiversity at the Universidad Nacional Mayor de San Marcos (UNMSM).

Surface Sterilization and DNA Extraction

To specifically assess the endophytic microbiota, all branch samples (AA, AI, AS) were re-sterilized in the laboratory (1% NaOCl, followed by three rinses in sterile water; 60 s each) to remove epiphytic bacteria. Internal tissues were then dissected following the protocol described in [14]. The three branch samples belonging to the same tree were pooled and ground together in liquid nitrogen, and approximately 100 mg of homogenized tissue was used for total DNA extraction using the DNeasy Plant Pro Kit (Qiagen, Germany), following the manufacturer's protocol.

For larval samples, approximately 15 mg of *Enallodiplosis discordis* larvae were pooled in a 1.5 mL tube. To minimize external DNA contamination, we followed an optimized larval pre-treatment protocol for insect endophytic/endosymbiont analyses: larvae underwent three 1 min washes in 70% ethanol, followed by three 1 min rinses in sterile distilled water [15]. This cycle was repeated twice. DNA was extracted using the ZymoBIOMICS DNA Miniprep Kit (Zymo Research, USA) according to the manufacturer's protocol.

DNA integrity was assessed by electrophoresis on a 1% agarose gel and visualized under UV light. DNA concentration and purity were determined spectrophotometrically

using a NanoDrop Lite (Thermo Scientific), with the 260/280 absorbance ratio used as an indicator of DNA purity.

16 S rRNA Library Preparation and Sequencing

Two libraries were prepared using the Oxford Nanopore 16 S Barcoding Kit 1–24 (SQK-16S024), which includes barcoded primers (27 F: AGAGTTTGATCCTGGCTCAG; 1492R: TACGGYTACCTTGTACGACTT) for 16 S rRNA gene amplification for up to 24 samples to enable subsequent demultiplexing. PCR reactions were performed under the following conditions to increase sequencing yield: reactions were carried out in a final volume of 40 μ L per sample, consisting of 25 μ L of LongAmp Hot Start Taq 2X Master Mix (New England Biolabs), 10 ng of template DNA, and 11 μ L of nuclease-free water. Thermocycling conditions were as follows: initial denaturation at 95 $^{\circ}$ C for 1 min; 35 cycles of 95 $^{\circ}$ C for 20 s, 55 $^{\circ}$ C for 30 s, and 65 $^{\circ}$ C for 2 min; and a final extension at 65 $^{\circ}$ C for 5 min.

PCR products were purified using AMPure XP magnetic beads (Beckman Coulter) according to the manufacturer's protocol and resuspended in 10 μ L of elution buffer (10 mM Tris-HCl, pH 8.0; 50 mM NaCl). Purified amplicons were normalized to equimolar concentrations using Tris-HCl/NaCl buffer, and 10 μ L aliquots from each sample were pooled. Subsequently, 1 μ L of Rapid Adapter (RAP) was added to the pooled library and incubated for 5 min prior to loading onto a MinION Mk1C sequencer (Oxford Nanopore Technologies).

Sequencing was performed using R9.4.1 flongle flow cells (FLO-MIN106). The flow cell was primed, and the sequencing mixture—comprising the sequencing buffer, loading beads, and the pooled DNA library—was loaded onto the flongle. This mixture stabilizes DNA strands and facilitates their translocation through the nanopores. Sequencing of the 16 S rRNA gene was carried out for approximately 12–18 h under default operating conditions. No mock community control was sequenced.

Bioinformatic Analysis

Sequencing data were first converted to POD5 format using POD5 v0.3.12 (<https://github.com/nanoporetech/pod5-file-format>) and subsequently basecalled and demultiplexed with Dorado v0.7.2 (<https://github.com/nanoporetech/dorado>). Adapter sequences (Rapid Adapters, RAP) were trimmed using Porechop v0.2.4 (<https://github.com/rrwick/Porechop>), and reads were then filtered by length and quality with Chopper v0.8.0 (<https://github.com/wdecoester/chopper>). Potential plant-derived contaminants were identified by aligning reads against the *N. pallida* genome (sequenced in-house) [16]. For larval datasets (ED), no explicit host-read

subtraction was applied, as no reference genome or closely related assembly is currently available for *E. discordis*. File handling was performed using Samtools v1.13 [17] and Bedtools v2.30.0 [18]. Quality-control metrics were calculated with SeqKit v2.9.0 [19] before and after filtering.

Taxonomic classification of filtered reads was performed using the NanoCLUST pipeline [17]. NanoCLUST groups similar reads using dimensionality reduction and unsupervised clustering, generates a polished consensus sequence for each cluster, and classifies these sequences using

Table 1 Number of sequences obtained after nanoclust classification, number of reads for each sample group, sample and sample replicate; and number of clusters per replicate for samples from asymptomatic trees, incipient diseased trees and late-stage symptomatic trees, and *E. discordis* larvae

Sample group	Sample	Sample replicate	Classification with NanoCLUST		
			N°. of reads	Reads per sample	N°. of clusters
Asymptomatic trees (AA)	AA-01	AA-01.1	1084	4592	2
		AA-01.2	2044		3
		AA-01.3	1464		3
	AA-02	AA-02.1	1180	3728	4
		AA-02.2	848		3
		AA-02.3	1700		3
	AA-03	AA-03.1	2232	5176	4
		AA-03.2	1532		4
		AA-03.3	1412		2
Late-stage symptomatic trees (AS)	AS-01	AS-01.1	704	3820	3
		AS-01.2	1388		3
		AS-01.3	1728		3
	AS-02	AS-02.1	1336	5420	3
		AS-02.2	2188		3
		AS-02.3	1896		3
	AS-03	AS-03.1	1824	5684	3
		AS-03.2	2092		3
		AS-03.3	1768		3
Incipient diseased trees (AI)	AI-01	AI-01.1	1850	11,066	5
		AI-01.2	884		3
		AI-01.3	8332		3
	AI-02	AI-02.1	6506	9398	3
		AI-02.2	354		3
		AI-02.3	2538		4
	AI-03	AI-03.1	9714	11,106	3
		AI-03.2	338		3
		AI-03.3	1054		4
<i>E. discordis</i> larvae (ED)	ED-01	ED-01.1	5240	12,656	8
		ED-01.2	3420		3
		ED-01.3	3996		32
	ED-02	ED-02.1	4044	12,772	17
		ED-02.2	4332		2
		ED-02.3	4396		10
	ED-03	ED-03.1	4284	14,796	33
		ED-03.2	5820		2
		ED-03.3	4692		5

BLAST against the updated NCBI 16 S rRNA database [20]. Descriptive and comparative analyses, as well as data visualization, were conducted in Python v3.10.12 via Google Colab, using libraries including pandas, numpy, matplotlib, seaborn, and scikit-learn. Outputs included bacterial composition profiles, alpha diversity metrics (Shannon, Simpson, and Chao1 indices), principal coordinates analysis (PCoA), and heatmaps based on the relative abundance of taxa at the genus and family levels. Although the aim of this study was to reveal trends at the descriptive level, formal statistical testing between sample groups was performed using R and included non-parametric methods PERMANOVA based on Bray-Curtis dissimilarities with 999 permutations for comparing bacterial compositions, and Kruskal-Wallis for evaluating statistically significant differences in alpha diversity metrics.

Results

Sequencing Success and Read Classification

A total of 28,420 reads were obtained from bacterial endophyte samples derived from both symptomatic and asymptomatic trees. In addition, 71,794 reads were generated from the endomicrobiome of *E. discordis* larvae and from endophytes of incipiently diseased (AI) trees associated with this insect. All raw sequences were deposited in the Sequence Read Archive (SRA) under BioProject PRJNA1314707. After NanoCLUST classification, read depth per group was 13,496 reads for AA, 31,570 for AI, 14,924 for AS, and 40,224 for (ED) (Table 1). Taxonomic assignments were successfully obtained for all samples.

Composition of Bacterial Endophytes

At the family level (Fig. 2A), asymptomatic trees (AA) were dominated by Symphyonemataceae (75.8–80.3%), with Stellaceae as the secondary family (≈ 10.7 –24.2%). However, as detailed below and discussed later, these assignments predominantly reflect plant organellar (chloroplast) DNA rather than true bacterial endophytes. Additional low-abundance families ($\leq \sim 10\%$) were detected sporadically, including Acidithiobacillaceae, Erwiniaceae, and Caedimonadaceae. In late-stage symptomatic trees (AS), Symphyonemataceae remained the most abundant family (60.2–80.9%), while Stellaceae increased in two samples (≈ 38 –40%). These patterns indicate that the endophytic microbiota of *N. pallida* stems is relatively simple and remains broadly similar between asymptomatic (AA) and late-stage symptomatic (AS) trees, suggesting that tree health status does not markedly affect the dominant

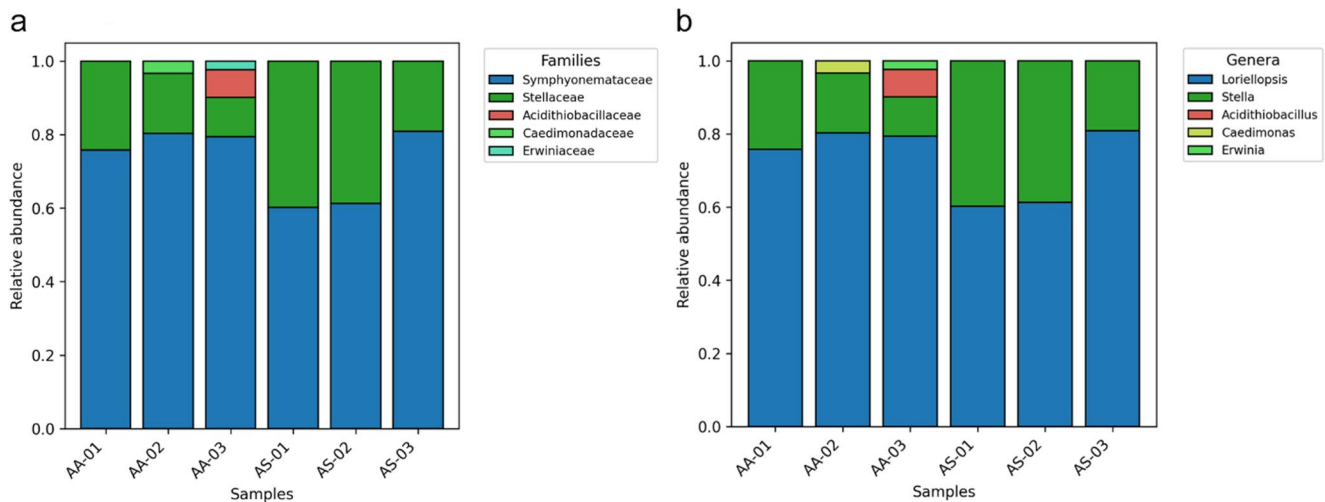


Fig. 2 Bacterial microbiota composition of endophytes from branches of asymptomatic and symptomatic *N. pallida* trees at the family level (a) and genus level (b). AA: Asymptomatic samples, AS: Late-stage symptomatic samples

bacterial families detected in woody tissues. Although this study was designed to describe compositional patterns, and the small sample size and number of taxa limits statistical, the non-parametric PERMANOVA test based on Bray-Curtis dissimilarities with 999 permutations showed no significant differences between any sample group (AA, AS, AI, ED) at the family level (pseudo-F: 0.47, $p = 1.00$; 999 permutations).

At the genus level (Fig. 2B), *Loriellopsis* predominated in asymptomatic trees (AA; 75.8–80.3%), with *Stella* as the secondary genus (≈ 10 –24%), both of which were predominantly organellar. Low-abundance genera ($< 8\%$), such as *Acidithiobacillus*, *Caedimonas*, and *Erwinia*, were detected in specific samples. Late-stage symptomatic trees (AS) exhibited a similar pattern, with *Loriellopsis* remaining dominant (60.2–80.9%) and *Stella* occurring at relatively higher proportions in two trees (≈ 38 –40%), while other genera were present at lower abundances. Similar to what was observed for the differences at the family level, bacterial composition was not significantly different among the four sample groups at the genus level (pseudo-F: 0.68, $p = 1.00$).

Comparison between the Endomicrobiome of *E. discordis* and Endophytes of *N. pallida*

At the family level (Fig. 3A), incipiently diseased (AI) branches were dominated by Symphyonemataceae (78.9–88.3%), with Stellaceae as the secondary family (≈ 10.7 –20.2%), whereas all other families occurred at lower proportions. In contrast, *E. discordis* larvae (ED) exhibited higher family-level diversity, with dominant families varying among samples. Across ED samples, Bacillaceae, Caryophanaceae, Acidithiobacillaceae, Aurantimonadaceae, and

Symphyonemataceae each reached approximately 14–44% in at least one sample (Fig. 3A), along with several additional low-abundance families.

At the genus level (Fig. 3B), incipiently diseased (AI) branches were characterized by a strong dominance of *Loriellopsis* (78.9–88.4%), with *Stella* as the secondary genus (≈ 10.7 –20.2%); all other genera were present at minor proportions (each $\leq 2\%$). In contrast, *E. discordis* larvae (ED) displayed heterogeneous microbial profiles, in which *Bacillus*, *Planococcus*, *Planomicrobium*, *Aureimonas*, *Acidithiobacillus*, and *Loriellopsis* each reached peaks of approximately 12–19% in at least one sample, with additional low-abundance genera illustrated in Fig. 3B.

These findings indicate that, whereas the microbiota of incipiently diseased *N. pallida* branches is overwhelmingly dominated by *Loriellopsis* with limited variation, *E. discordis* larvae harbor a more complex and diverse microbial community comprising a broader array of bacterial genera. Our results suggest that larvae acquire part of the tree-associated microbiota but also harbor additional bacterial groups that are likely linked to their physiology, digestive processes, or microhabitat within plant tissues.

Diversity Analysis

Alpha diversity was consistently higher in *E. discordis* larvae than in tree-associated samples across the dataset. Shannon diversity values were highest in the ED group, whereas AA, AI, and AS clustered at lower values (Fig. 4A). Similarly, Simpson indices indicated greater evenness in ED samples (Fig. 4B), and Chao1 estimates likewise suggested higher richness potential in ED compared with any tree-associated group (Fig. 4C). Although these trends were clear at the descriptive level, statistical testing revealed

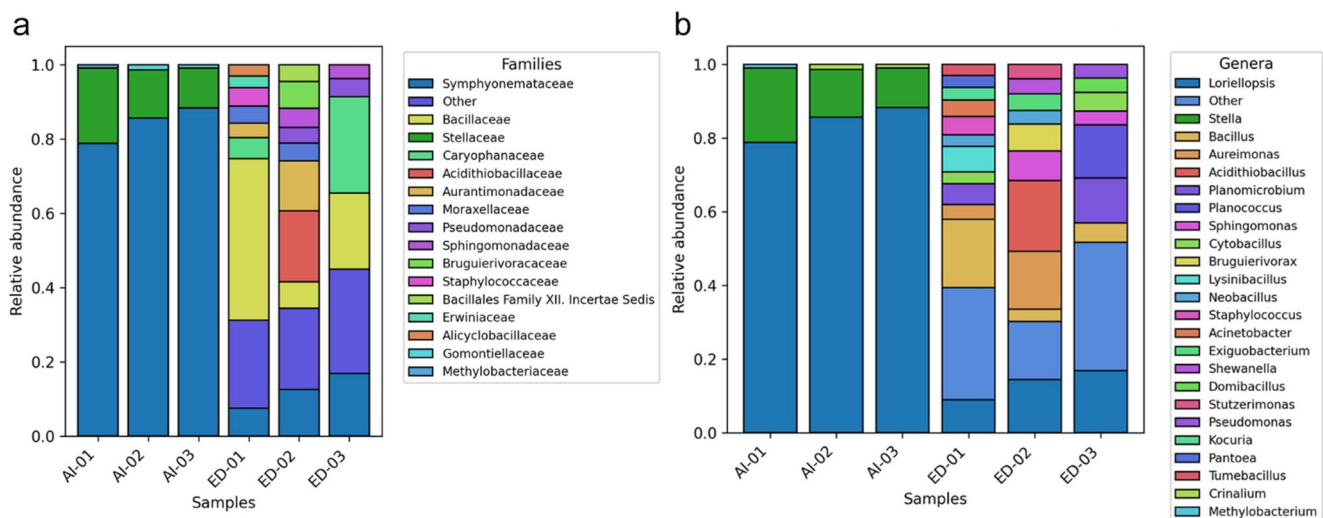


Fig. 3 Bacterial microbiota composition of the endomicrobiome of *E. discordis* and of the endophytes from branches of incipient diseased *N. palida* trees infested with *E. discordis* at the family (a) and genus (b) levels

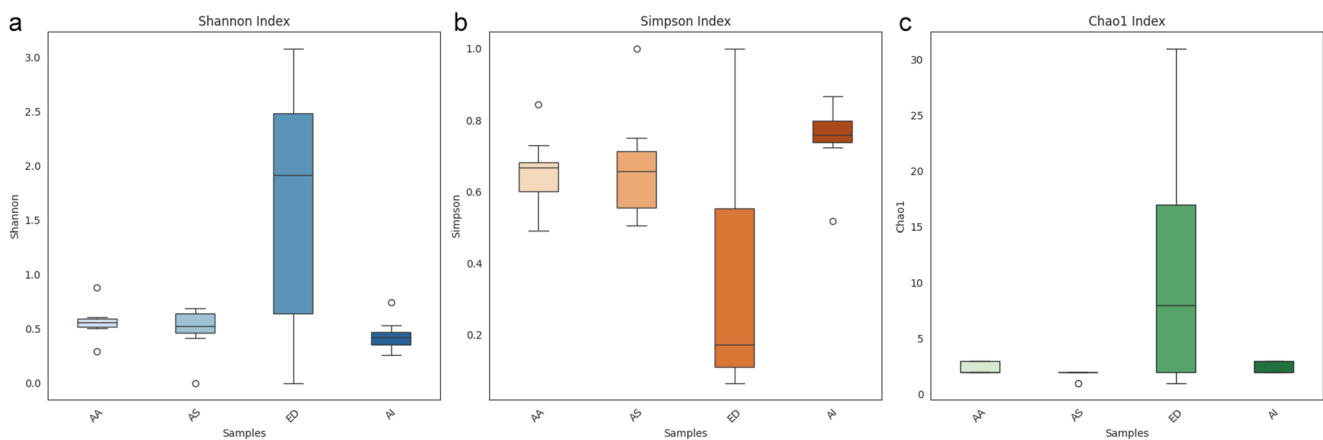


Fig. 4 Assessment of alpha diversity in the different groups analyzed using the Shannon (a), Simpson (b), and Chao1 (c) indices

significant differences only for the Simpson index (Kruskal-Wallis test; $H = 8.27$, $p = 0.0407$), while Shannon ($H = 7.45$, $p = 0.0587$) and Chao1 ($H = 4.90$, $p = 0.1790$) diversity indices did not differ significantly among sample groups, likely reflecting the high intra-group variability observed in ED samples. Collectively, these patterns indicate a broader and potentially more functionally flexible gut microbial community in larvae, whereas branch endophytes reflect host-filtered, lower-diversity assemblages in this dataset. Motivated by these within-sample differences, we next evaluated between-sample structure using principal coordinates analysis (PCoA).

PCoA Analysis

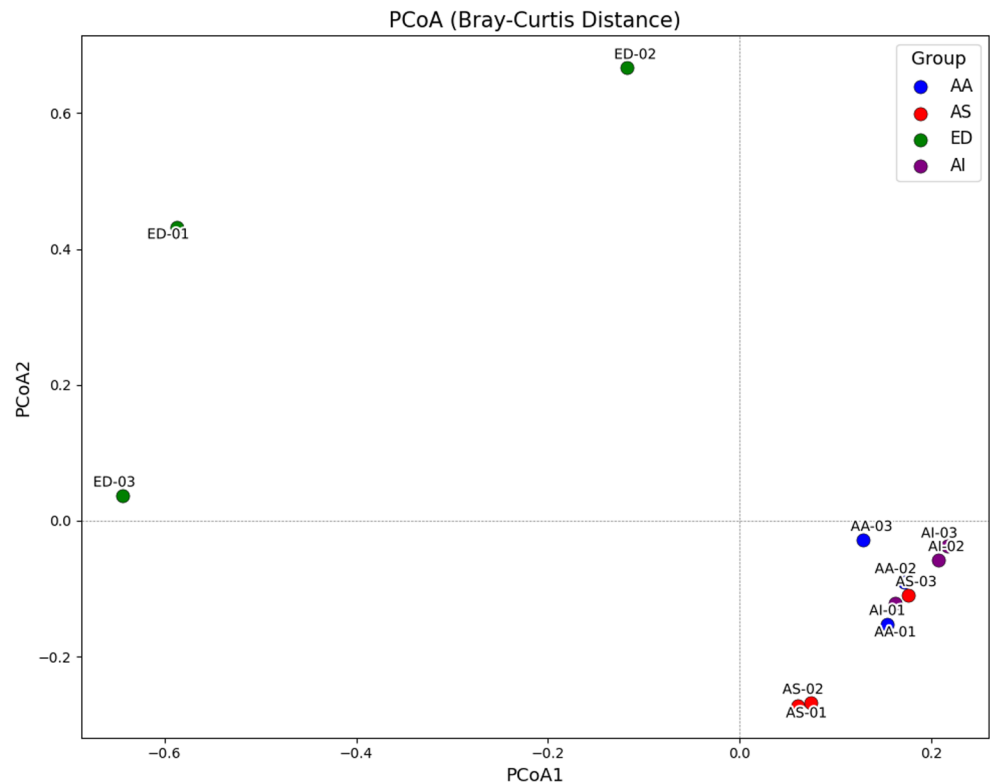
PCoA revealed a clear separation between larval (ED) and branch samples (AA, AI, AS) (Fig. 5), suggesting that the endophytic microbiota of branches remains relatively stable

despite differences in plant health status. Nonetheless, some variation was observed among AS-01 and AS-02 samples, attributable to differences in the relative abundance of specific genera, such as *Loriellopsis* and *Stella*. In contrast, *E. discordis* larvae (ED) exhibited a completely distinct and highly variable microbiota in our dataset (Fig. 5), indicating a community adapted to the larval lifestyle and potentially changing according to developmental stage. This marked difference in microbiome diversity between branches and larvae is consistent with the contrasting nature of the microhabitats represented by woody tissues and insect larvae. To identify the taxa underlying this contrasting pattern, we next examined genus-level heatmaps.

Heatmaps of Microbial Taxa Correlations

To explore patterns among bacterial taxa within each group, heatmaps were generated to display the relative abundance

Fig. 5 PCoA analysis of the different samples from *N. pallida*: asymptomatic tree, incipient diseased tree and late-stage diseased tree, and of the larvae from incipient diseased trees



(percentage) of bacterial genera across individual samples (Fig. 6). Higher relative abundance values are represented by lighter colors (yellow and green), whereas low or absent values are shown in darker tones.

The genus *Loriellopsis* exhibited high relative abundance across all sample groups (AA, AI, AS, and ED), exceeding 85% in certain samples. This pattern suggests that *Loriellopsis* is a common endophyte of *N. pallida* and that *E. discordis* larvae may acquire it through feeding on leaflets. In contrast, the genus *Stella* was detected exclusively in tree-associated samples (AA, AI, and AS), with a stronger presence in branch tissues. This finding indicates that *Stella* is a branch-associated bacterial endophyte of *N. pallida* and is absent from leaflets and from *E. discordis* larvae.

With respect to the microbiota associated with *E. discordis*, genera such as *Bacillus*, *Planococcus*, and *Aureimonas* were detected in the ED group, suggesting potential symbiotic associations with the insect host. In addition, although the genus *Acidithiobacillus* was detected in some samples, its intermediate relative abundance (below 20%) indicates that it is not a dominant component of the microbiota of either *N. pallida* or *E. discordis* and may represent a transient presence (Fig. 6). Similar patterns were observed when analyses were performed at the family level (Fig. S1). Taken together, these results indicate that, although *Loriellopsis* represents a shared component across hosts and tissues, the microbiota of *E. discordis* displays greater breadth and compositional differentiation, most likely driven by

insect-associated factors rather than by the tree environment alone.

Discussion

In branches from late-stage symptomatic trees (AS), the apparent bacterial signal was dominated by sequences matching plant organelles (chloroplasts and mitochondria), and genuine endophytic bacterial taxa were scarce in our dataset. By contrast, low-abundance bacterial families were detected in some asymptomatic (AA) and incipiently diseased (AI) branches, including Acidithiobacillaceae, Caedimonadaceae, and Erwiniaceae. *Acidithiobacillus* may be associated with residual mining activity in the Pómac Historical Sanctuary, as members of this genus are known to oxidize sulfur compounds and mobilize metals such as gold, copper, and uranium [21, 22]. The presence of *Caedimonas*, described as an endosymbiont of protozoa [23], could reflect environmental contamination or an as-yet undescribed ecological function. *Erwinia*, reported as a pathogen in guava and eucalyptus [24, 25], may act as an opportunistic pathogen under specific host physiological conditions.

Families Symphyonemataceae and Stellaceae accounted for much of the apparent signal across AA, AI, and AS branches. However, further alignment diagnostics suggested that these reads matched plant chloroplast and, to a lesser extent, mitochondrial genomes more closely than bona fide

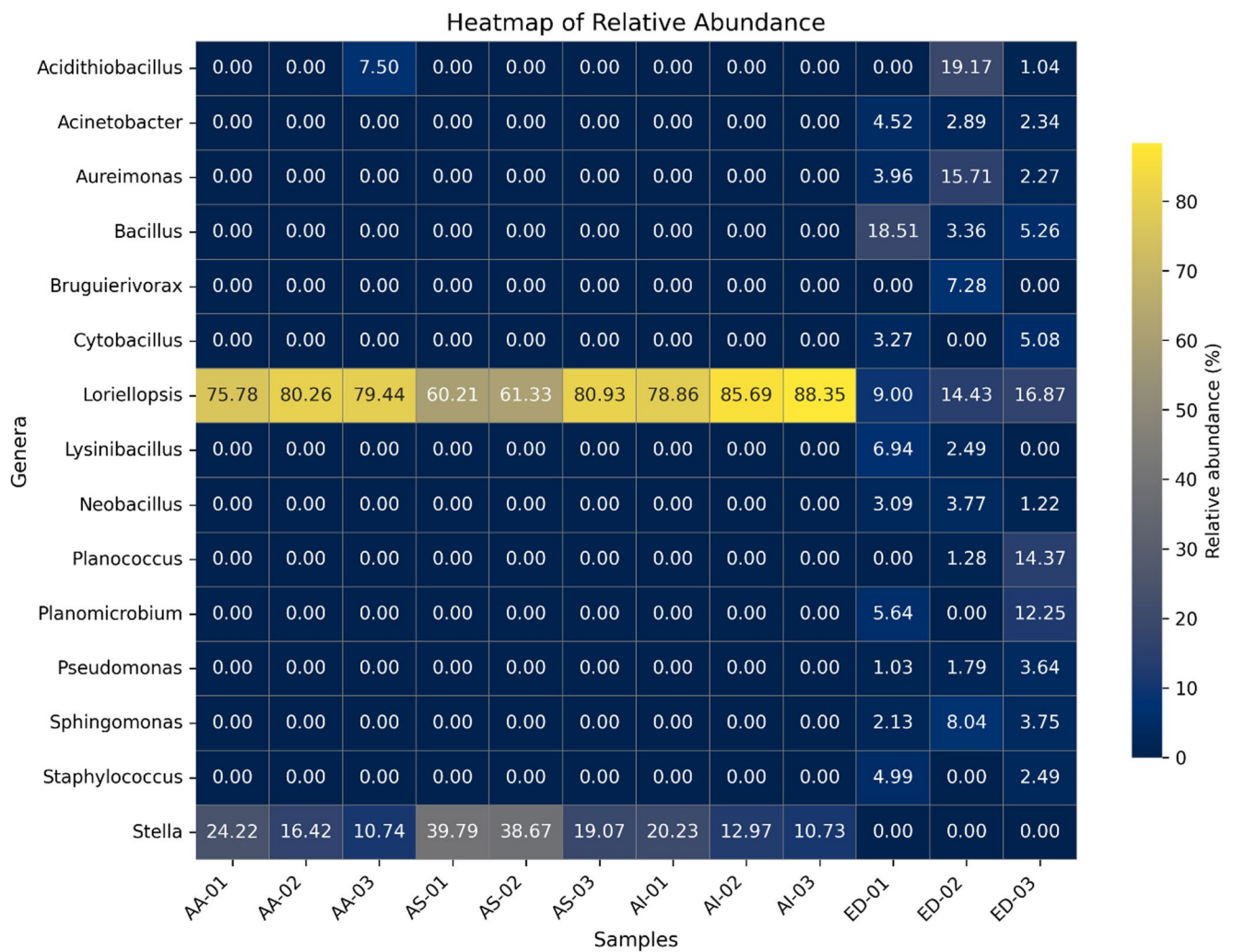


Fig. 6 Heatmap showing the percentage of bacterial genera in *E. discordis* samples and in *N. pallida* trees under different health conditions

bacterial references. We therefore interpret these assignments as organellar carryover rather than genuine endophytic cyanobacteria in our dataset. This interpretation is also consistent with the limited evidence for endophytic occurrences of these families; except, Stellaceae has been reported mainly from agricultural soils, with proposed roles in phytoremediation and cellulose degradation [26]. Given the homology between chloroplast loci and bacterial 16 S rRNA targets, aggressive host-read subtraction can introduce false negatives; accordingly, we retained these reads for completeness, interpreted them as organellar background, and analyzed genuine low-abundance bacterial families separately. The methodological limitations in identifying endophytes was reported before using 16 S primers, including the pair used in this study [27], requiring more specific primers or alternative methods [28, 29]. In our dataset, larval communities displayed greater richness and evenness than branch endophytes (Fig. 4) and occupied a broader ordination space (Fig. 5). Rather than being dominated by a single lineage, ED

samples harbored a multi-lineage assemblage encompassing families commonly reported from insect guts, including Acidithiobacillaceae, Aurantimonadaceae, Bacillaceae, Moraxellaceae, Pseudomonadaceae, Sphingomonadaceae, and Methylobacteriaceae [30–39]. Between-sample heterogeneity may also reflect differences in larval developmental stages [40]. This elevated diversity was accompanied by substantial intra-group variability among larval samples, which likely constrained the statistical detection of consistent differences across all alpha diversity metrics. The detection of Symphyonemataceae in *E. discordis* samples is most likely attributable to ingestion of *N. pallida* foliage; as noted above, sequences assigned to this family show greater similarity to chloroplast genomes than to genuine bacterial references. Thus, their presence in the insect gut likely reflects dietary intake rather than an endosymbiotic association.

At the genus level, the larval microbiome was more consistent with insect-associated metabolism than with tree endophytes. *Bacillus*—recurrently reported in agricultural

pests such as *Hermetia illucens* and *Rhynchophorus ferrugineus*—supports cellulose degradation and contributes to immunity and detoxification processes [33, 34, 39, 41–43]. *Pseudomonas*, documented in *Nilaparvata lugens* and *Tuta absoluta* and also detected in *Gynaikothrips uzeli* and adult *Liometopum apiculatum*, is consistent with facultative symbiosis and the processing of complex polysaccharides and aromatic compounds [44–47]. Complementing these functions, *Sphingomonas* participates in the turnover of recalcitrant substrates and has been linked to nitrogen cycling in *L. apiculatum* [44, 47]. *Acinetobacter* aligns with insecticide degradation (e.g., in *T. absoluta*) [44, 48], whereas *Aureimonas* has been associated with gut responses to insecticides in *Bombyx mori* and *Piezodorus guildinii* [28, 29]. Finally, *Methylobacterium* has been implicated in amino acid metabolism and digestive enzyme synthesis in insects [15, 39]. Taken together, these genera suggest an intestinal community configured for cellulose depolymerization, xenobiotic detoxification, and nutrient acquisition, consistent with the broad and heterogeneous profiles observed in *E. discordis* larvae. Comparisons of relative bacterial abundance between *N. pallida* endophytes and the gut microbiota of *E. discordis* should be interpreted with caution, as these communities occur in distinct biological kingdoms and ecological contexts. Taxa that are abundant or potentially beneficial in the tree may therefore occur at low relative abundance in the insect host without implying limited functional relevance. Accordingly, host–insect microbiome comparisons in this study are presented descriptively and do not allow inference of microbial transfer based on relative abundance alone.

Overall, branch endophytes represent a restricted community dominated by Symphyonemataceae/*Loriellopsis*, which we interpret as organellar carryover rather than true bacterial endophytes. In contrast, *E. discordis* larvae harbors a broader and more heterogeneous assemblage of taxa that partially overlaps with branch-associated sequences but also includes bacterial lineages enriched in insect-associated functions. In this context, no pattern consistent with systemic bacterial dysbiosis was observed in the branches of *N. pallida*. Consequently, the hypothesis that *E. discordis* acts as a vector of systemic phytopathogenic bacteria is not supported by our results. Instead, it is more plausible that the observed syndrome is localized in foliar tissues or involves other microorganisms (e.g., fungi or viruses) acting under larval feeding stress. This interpretation is consistent with acoustic tomography data taken previously (C. Arbizu, PhD, Director de la Subdirección de Investigación y de Estudios Especiales, Instituto Nacional de Innovación Agraria-INIA, email, April 29, 2022) confirming internal trunk integrity in symptomatic trees prior to the onset of this study (Fig. S2). Nevertheless, a larger sample size and/or greater sequencing depth would be valuable to further test this hypothesis. In addition, classical

microbiological isolation of bacteria from necrotic or chlorotic leaf tissues, followed by 16 S rRNA gene sequencing of individual colonies, could help assess microbial viability and clarify any potential involvement in defoliation. Finally, the implementation of Nanopore Adaptive Sampling (NAS) or the use of more selective 16 S rRNA primer sets designed to minimize chloroplast amplification would likely enhance the detection of low-abundance bacterial taxa that may otherwise remain undetected.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s00248-026-02711-8>.

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Author Contributions Gisella Orjeda conceived and designed the study. Material preparation, data collection and analysis were performed by Rodrigo Suarez, Esteban Caycho, Manuel Saucedo and Gisella Orjeda. Bioinformatic analysis pipeline was performed by Rodrigo Suarez, Manuel Ramirez and Renato La Torre. The first draft of the manuscript was written by Rodrigo Suarez and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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Data Availability All raw sequences were submitted as SRA under Bioproject PRJNA1314707.

Declarations

Competing interests The authors declare no competing interests.

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